

I vs. Me : An adversarial interaction with your past self

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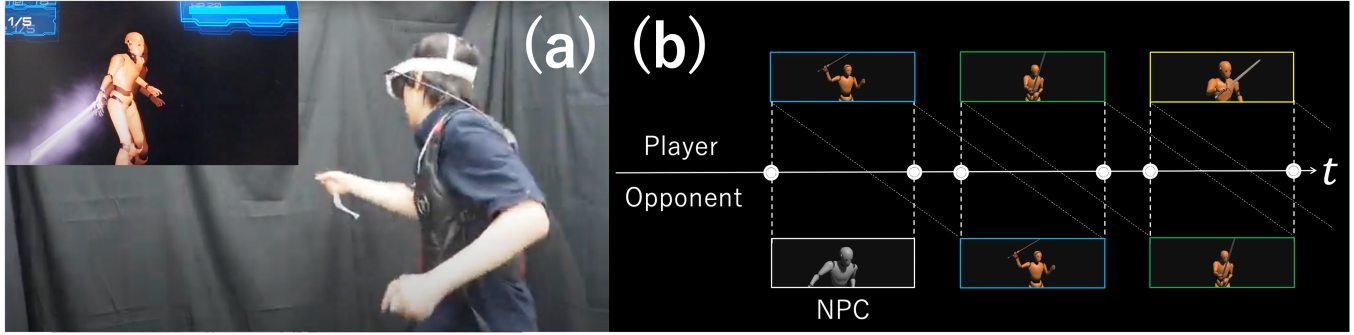


Figure 1: (a) A player of I vs. Me who has a sword fight with his past self. (b) The interaction process of I vs. Me.

Abstract

Interaction with a virtual self has been the focus of interest due to its psychological effects. In order to explore new possibilities for interaction with a virtual self, we propose an adversarial interaction with the past self. The player of our work interacts with a virtual self who reproduces the movements that the player experienced a few dozen seconds ago. However, because the player does not clearly remember the movements, they have to respond immediately to the opponent's attacks, which are difficult to predict. Here is how the process works; the player initially fights with a non-player character (NPC). The player's movements during this fight are recorded and become the movements of the opponent in the next round. After that, the player's movements in the current round become the movements of the opponent in the next round.

This research will advance the understanding of the psychological effects of interacting with a virtual self, and it is also expected to be applied to individual training methods in interpersonal sports.

CCS Concepts

• **Human-centered computing** → *Interaction design theory, concepts and paradigms.*

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1 INTRODUCTION

Interaction with a virtual self has been the focus of interest in both academic and artistic fields. As a virtual self-similar existence that behaves autonomously independent of the user it was modeled after, the concept of a virtual doppelganger has been proposed [Bailenson and Segovia 2010]. Interactions with virtual doppelgangers are studied with a focus on psychological effects [Guo et al. 2024]. In contrast, an example of interaction with a non-autonomous self-image is “Through the Looking Glass” [Kakehi and Naemura 2005]. In this art piece, the player alternately controls the physical self in the real world and the reflection of the self in the mirror, engaging in a game of air hockey. The player is always also an opponent, and the player does not perform simultaneous actions in response to the opponent's actions.

The past self does not behave autonomously, independent of the self, but it can behave differently from the “current” self. In particular, a slightly earlier past self is an interesting interaction partner that is somewhere in between a virtual doppelganger and the reflection of the self in the mirror, because the current self has

memories that guarantee non-independence of the past self from the self.

In order to explore new possibilities for interaction with a virtual self, we propose an adversarial interaction with the past self.

An example of an interaction with the past self is the game “We are one” [Flat Head Studio 2023]. In this game, you create multiple clones of your past self by recording your past movements multiple times, and then you achieve the goal by cooperating with your clones. In this game, remembering your past actions is an important element for achieving effective cooperation.

Interestingly, in the adversarial interaction experience we propose, the fact that you can’t remember the details of your own actions from a few dozen seconds ago plays an important role. This is in contrast to the case of cooperating with your past self. The player of our work interacts with a virtual self who reproduces the movements that the player experienced a few dozen seconds ago. However, because the player does not clearly remember the movements, they have to respond immediately to the opponent’s attacks, which are difficult to predict.

2 INTERACTION DESIGN

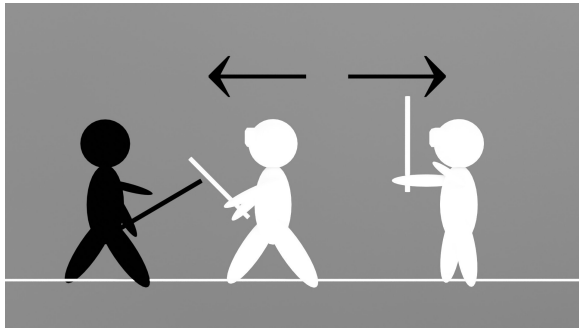


Figure 2: One-dimensional movements in the sword fight.

In this work, the player wears a head-mounted display (HMD) and engages in a sword fight with the opponent who reproduces the player’s past movements. We used avatars¹ and swords² sourced from the Unity Asset Store.

2.1 Interaction Process

Figure 1.b shows how the process works: the player initially fights with a non-player character (NPC). The player’s physical movements during this fight are recorded and become the movements of the opponent in the next round. After that, the player’s movements in the current round become the movements of the opponent in the next round, and this process is continuously repeated.

By setting the length of each round appropriately, the player can fight with their past self with the memory that they are not independent of the current self. Additionally, in this process, the player’s current movements are not reflected in their opponent’s

movements while one round. This helps the player to concentrate on the fight with the opponent without trying to memorize their current movements.

2.2 The Rules of The Sword Fight

The rules of sword fighting are also important in order to guarantee an adversarial interaction. In this experience, the player has a sword and is restricted to moving in one-dimension, forward and backward (Figure 2). Therefore, the attacks of the opponent will always be effective attacks that face the player even though the opponent’s movements only replicate the player’s past actions. Furthermore, the player has Hit Points (HP), requiring them to avoid attacks from the opponent. As a result, the player will carefully watch the opponent’s movements, deciding whether to move forward to attack or backward to evade the opponent’s attack.

3 USER EXPERIENCE

This work was exhibited at IVRC2024 in Nagoya, Japan, from September 11 to 13, 2024 (Figure 1.a). Participants who experienced “I vs. Me” provided feedback such as, “It is indeed difficult to recall my movements from the previous round when facing an opponent right in front of me.” Many of those participants acknowledged that the experience of fighting with their past selves was well executed.

They described their opponent avatar as “a being with the same thoughts as myself,” or “an uncooperative myself that was frustrating to me,” among other descriptions. However, it should be noted that: at this time, the appearance of the participant was not reflected in the avatar.

4 DISCUSSION AND FUTURE WORK

Our system aims to realize an adversarial interaction with the past self. In this system, the player’s past movements are reproduced as the actions of the opponent avatar. However, during the workshop described in Section 3, some participants commented that “the opponent’s movements did not feel like my own.” This recognition is noteworthy.

In the future, we will focus on the cognitive and psychological aspects of the players. We will investigate how their perception and behavior change under various conditions, such as when the opponent’s movements are partially or fully autonomous, or when the avatar’s appearance is self-similar or different. We also plan to consider applying this system to individual training methods in interpersonal sports such as fencing.

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¹Avatar: Crash Test Dummy (Color Customizable) from Unity Assets Store, <https://assetstore.unity.com/packages/3d/characters/crash-test-dummy-color-customizable-227040>, Accessed: 2024-09-28

²Sword: Laser Sword PRO from Unity Assets Store, <https://assetstore.unity.com/packages/tools/particles-effects/laser-sword-pro-93927>, Accessed: 2024-09-28